

Fall 2024

Noetic Learning Mathematics Contest

Grade 3



45 minutes • No calculators allowed

Student Name: _____

For Teacher Use Only

Number of correct answers: _____

Score (Number of correct answers $\times$ 5): _____

1) $20 + 24 = ?$

Answer: _____

- 2) Chloe is celebrating her 15th birthday. Nine years ago today, her mother gave her a newborn puppy named Coco as a birthday gift. How many years older is Chloe than Coco?

Answer: _____ years older

- 3) Trevor uses sticks of equal length to build triangles for his ladybugs. Each side of a triangle is made with 1 stick, and all triangles are built flat on the ground. He needs 3 sticks to build 1 triangle and 5 sticks to build 2 triangles, as shown below. What is the minimum number of sticks he needs to build 4 triangles for his 4 ladybugs?



Answer: _____ sticks

- 4) Sanya has 3 trees in her backyard. At the beginning of October, each tree had 60 leaves. During the month of October, each tree lost half of its leaves. What is the total number of leaves that were left on all 3 trees at the end of October?



Answer: _____ leaves

- 5) At a carnival, each ride ticket costs \$3. At most, how many ride tickets can Barbi buy with \$20?

Answer: _____ ride tickets

- 6) Jake is a barber, and he has to give 4 haircuts today. Each haircut lasts 25 minutes. Jake takes a 10-minute break after each haircut, and he doesn't take any other breaks. How many minutes does it take from the start of his first haircut to the end of his last haircut?

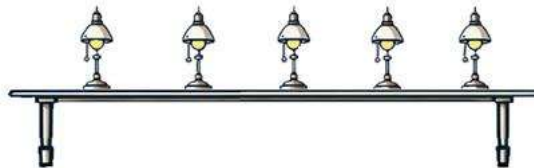
Answer: _____ minutes

- 7) Today is Wednesday, and Stella's 18th birthday party is this Saturday. Starting today, Stella's mom makes 5 paper flowers each day before Saturday. How many more paper flowers does Stella's mom need to make on Saturday to have a total of 18 paper flowers?



Answer: _____ more paper flowers

- 8) There are 5 lamps on a library table, and they are all turned on. Each lamp has a chain that can be pulled to turn it on and off. Pulling the chain once turns the lamp on if it is off and turns the lamp off if it is on. Ray pulls the chain of the first lamp 1 time, the second lamp 2 times, the third lamp 3 times, the fourth lamp 4 times, and the fifth lamp 5 times. How many lamps will be on after Ray is done?



Answer: _____ lamps

- 9) Lila goes strawberry picking. She eats every third strawberry she picks (the 3rd, 6th, 9th, and so on) and puts the others in a basket. Lila picks a total of 20 strawberries. How many strawberries does she put in the basket?

Answer: _____ strawberries

- 10)** Four people, labeled A, B, C, and D, each hold two cards with the same number. Some cards are held backwards, so the numbers are not visible. The sum of all eight numbers is 52. What number is on each of the cards that person C is holding?



Answer: _____

- 11)** The width of a rectangular pool is 7 yards, and its perimeter is 36 yards. What is the length of the pool?

Answer: _____ yards

- 12)** In 10 minutes, it will be 8:15 AM. Bill left for school 15 minutes ago. How many minutes after 7:00 AM did Bill leave for school?

Answer: _____ minutes

- 13)** In the subtraction problem below, place the numbers 1, 2, 3, 4, and 5 into the five squares so that the difference is as large as possible. Use each number only once. What is the largest possible difference?

$$\begin{array}{r} \square \square \square \\ - \quad \square \square \\ \hline \end{array}$$

Answer: _____

- 14)** Aiya bought an apple, a grapefruit, and a pumpkin. The apple and the grapefruit together weigh 25 ounces. The apple and the pumpkin together weigh 59 ounces. The pumpkin weighs 52 ounces. How many ounces does the grapefruit weigh?

Answer: _____ ounces

- 15)** A vet saw a total of 18 cats and dogs today. He saw 4 more cats than dogs. How many cats did he see today?

Answer: _____ cats

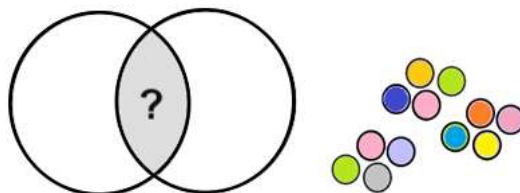
- 16)** There are a total of 15 pens in a drawer: 7 blue pens and 8 red pens. Without looking, what is the smallest number of pens you need to take out of the drawer to be sure that you take out at least 2 blue pens?

Answer: _____ pens

- 17)** David is waiting in line to ride the roller coaster. He is the 15th person when counting from the front of the line, and he is also the 15th person when counting from the back of the line. Ethan is waiting in the same line, and he is the 8th person from the front. How many people are in line behind Ethan?

Answer: _____ people

- 18)** Jack has 12 marbles. He places all 12 marbles inside the two circles below so that each circle has exactly 9 marbles inside. How many marbles does Jack place inside the shaded part, where the two circles overlap?



Answer: _____ marbles

- 19)** In a class of 18 students, each student raised either 1 hand, 2 hands, or no hands. Mrs. Nathan counted a total of 21 hands raised. She also counted that 3 students did not raise any hands. How many students raised 2 hands?

Answer: _____ students

- 20)** Six people want to cross a river. They only have one small boat that can carry at most 3 people at a time. One person must return with the boat after each trip to pick up more people. It takes 3 minutes for the boat to cross the river from one side to the other. What is the fewest number of minutes it will take for all 6 people to cross the river? (Note: You don't need to count the time it takes to get on and off the boat.)



Answer: _____ minutes

~ The End ~

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Grade 3 Answers & Solutions

1) (44) Solution: $20 + 24 = 44$

2) (6 years older) Solution: Nine years ago, Coco was a newborn, so Coco is now 9 years old. Chloe is 15 years old, so she is $15 - 9 = 6$ years older than Coco.

3) (9 sticks) Solution: Trevor needs 3 sticks to build 1 triangle and 5 sticks to build 2 triangles. He needs 2 more sticks to make the 3rd triangle, so he needs $5 + 2 = 7$ sticks for 3 triangles. He needs 2 more sticks to make the 4th triangle, so he needs a total of $7 + 2 = 9$ sticks for 4 triangles.



4) (90 leaves) Solution: At the beginning of October, each of Sanya's 3 trees had 60 leaves. During the month, each tree lost half of its leaves, so each tree has 30 leaves at the end of October. Since there are 3 trees, the total number of leaves remaining at the end of October is $30 + 30 + 30 = 90$ leaves.

5) (6 ride tickets) Solution: To find out how many tickets she can buy, we use a table to find the cost of 1 ticket, 2 tickets, 3 tickets, and so on, until we reach \$20:

Number of tickets	1	2	3	4	5	6	7
Cost	\$3	\$6	\$9	\$12	\$15	\$18	\$21

Since 7 tickets cost \$21 (more than \$20), Barbi can buy at most 6 tickets.

6) (130 minutes) Solution: Each of Jake's haircuts lasts 25 minutes, so the total time for 4 haircuts is $25 + 25 + 25 + 25 = 100$ minutes. He takes a 10-minute break after 1st haircut, 2nd haircut, 3rd haircut, so the total break time is $10 + 10 + 10 = 30$ minutes. In all, it takes Jake a total of $100 + 30 = 130$ minutes from the start of his first haircut to the end of his last haircut.

7) (3 more paper flowers) Solution: Stella's mom makes 5 paper flowers each day from Wednesday to Friday, for a total of $5 + 5 + 5 = 15$ flowers. To have a total of 18 flowers, she needs to make $18 - 15 = 3$ more flowers on Saturday.

8) (2 lamps) Solution: Each pull changes the state of the lamp from on to off or from off to on. So, 1 pull $\rightarrow$ off, 2 pulls $\rightarrow$ on, 3 pulls $\rightarrow$ off, 4 pulls $\rightarrow$ on, and 5 pulls $\rightarrow$ off.

Ray pulls the first lamp 1 time, the second lamp 2 times, the third lamp 3 times, the fourth lamp 4 times, and the fifth lamp 5 times. Therefore, the lamps that are on at the end are the 2nd lamp and the 4th lamp. So, there are 2 lamps on at the end.

9) (14 strawberries) Solution: Count by 3s to find out the strawberries Lila eats: 3, 6, 9, 12, 15, 18. So, she eats a total of 6 strawberries. The remaining $20 - 6 = 14$ strawberries she put in the basket.

10) (9) Solution: Since everyone holds two cards with the same number, we know the blank card that B holds is 7 and the blank card that D holds is 6.



Find the sum of the six cards: $4 + 4 + 7 + 7 + 6 + 6 = 34$. Because the sum of all 8 numbers is 52, the two numbers that C holds have a sum of $52 - 34 = 18$. Since these two numbers are the same, each number is half of 18, which is 9.

11) (11 yards) Solution: The perimeter of the pool consists of 2 widths and 2 lengths. Since the width is 7 yards, the total of the two widths is $7 + 7 = 14$ yards. Subtracting the two widths from the perimeter of the pool (36 yards), we get the total of the two lengths: $36 - 14 = 22$ yards. Each length is half of 22, which is 11 yards.

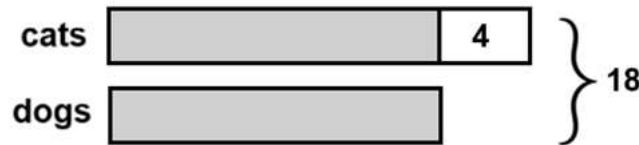
12) (50 minutes) Solution: In 10 minutes, it will be 8:15 AM, so the current time is 8:05 AM. Bill left for school 15 minutes earlier, so the time he left for school is 7:50 AM, which is 50 minutes past 7:00 AM.

13) (531) Solution: To get the largest difference, we need to place the largest possible 3-digit number on top and the smallest possible 2-digit number on the bottom. The largest possible 3-digit number from the given digits (1, 2, 3, 4, and 5) is 543, and the smallest possible 2-digit number is 12. Thus, the largest possible difference is $543 - 12 = 531$.

$$\begin{array}{r} \boxed{5} \boxed{4} \boxed{3} \\ - \quad \boxed{1} \boxed{2} \\ \hline \end{array}$$

14) (18 ounces) Solution: First, find the weight of the apple by subtracting the weight of the pumpkin (52 ounces) from the combined weight of the apple and pumpkin (59 ounces): $59 - 52 = 7$ ounces. Next, find the weight of the grapefruit by subtracting the weight of the apple (7 ounces) from the combined weight of the apple and grapefruit (25 ounces): $25 - 7 = 18$ ounces.

15) (11 cats) Solution: Draw a bar diagram with one bar representing the number of cats and another bar representing the number of dogs.

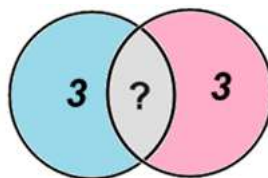


The total of the two bars is 18, and the bar representing cats is 4 units longer than the bar representing dogs. Subtract the extra 4 from 18: $18 - 4 = 14$. As shown in the diagram above, 14 represents the total of the two equal gray-shaded parts. Therefore, each gray-shaded part represents $\frac{1}{2}$ of 14, which is 7. Therefore, the vet saw 7 dogs and $7 + 4 = 11$ cats today.

16) (10 pens) Solution: Consider the worst-case scenario where you pick all 8 red pens first. After picking 8 red pens, the next 2 pens you take will both be blue. Therefore, you need to take out 10 pens to ensure that at least 2 are blue.

17) (21 people) Solution: David is the 15th person from both the front and the back, so there are 14 people before him and 14 people after him. So, there are $14 + 14 + 1$ (David) = 29 people total in line. Since Ethan is the 8th person from the front, the number of people behind Ethan is $29 - 8 = 21$.

18) (6 marbles) Solution: Since there are 12 marbles total and 9 marbles in the first circle, there are $12 - 9 = 3$ marbles outside the first circle. Place these 3 marbles in the pink-shaded region. Similarly, there are $12 - 9 = 3$ marbles outside the second circle. Place these 3 marbles in the blue-shaded region. The remaining $12 - 3 - 3 = 6$ marbles must be in the overlapping region.



19) (6 students) Solution: There are 18 students, and 3 students didn't raise any hands, so $18 - 3 = 15$ students raised either 1 hand or 2 hands. Since a total of 21 hands were raised, $21 - 15 = 6$ students raised both hands.

20) (15 minutes) Solution: Let's figure out how many crossings it takes:

	# of people on one side of the river	# of people on the other side of the river
After 1 st crossing	3	3
After 1 st return crossing	4	2
After 2 nd crossing	1	5
After 2 nd return crossing	2	4
After 3 rd crossing	0	6

It takes a total of 5 crossings to get all 6 people to the other side. Each crossing takes 3 minutes. So, it takes $3 \times 5 = 15$ minutes.

~ The End ~